

Homework 8 - Assignment Notes

The following document discusses various mathematical concepts and new MATLAB commands that you may need in Homework 8.

Equation of a line

Every line can be written in the form

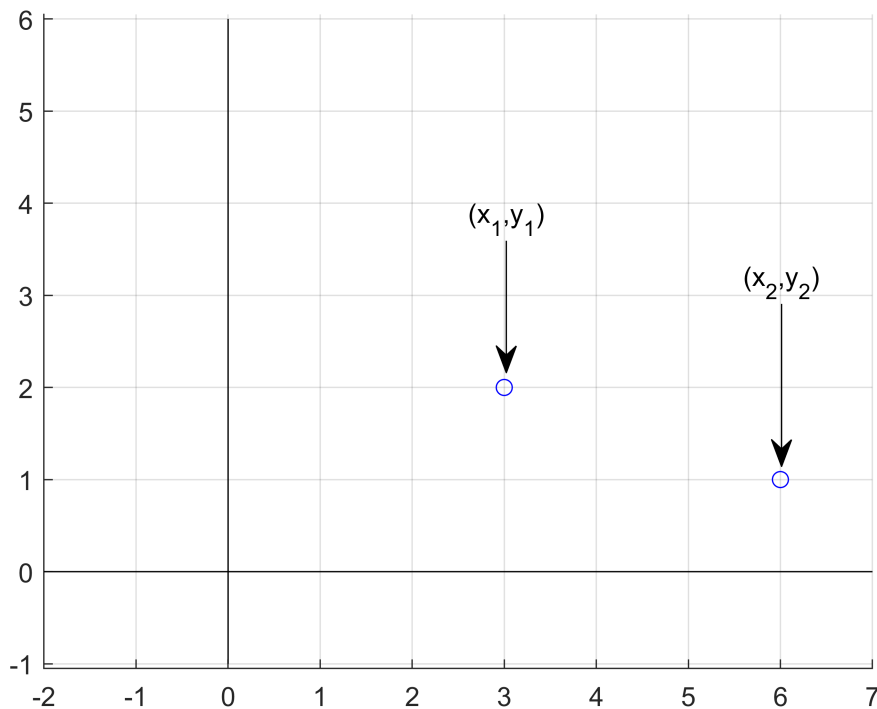
$$y = m x + b \quad (1)$$

where m is the slope and b is the y -intercept.

In this assignment, you will need to find the equation of a line that pass through two points. Here is a quick review:

EXAMPLE. Suppose you want to find the equation of a line that passes through $(3, 2)$ and $(6, 1)$.

Here is a plot of the two points:



First, compute the slope, m .

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1 - 2}{6 - 3} = \frac{-1}{3} = -\frac{1}{3}$$

Once, you have m , you can find b . Here is the idea:

Solve eqn (1) for b

$$b = y - m x$$

Now, plug in m (above) and one of the points on the line, say $(3, 2)$

$$b = y - m x = 2 - \left(-\frac{1}{3}\right)(3) = 2 + 1 = 3$$

Therefore, the eqn of the line is

$$y = -\frac{1}{3}x + 3$$

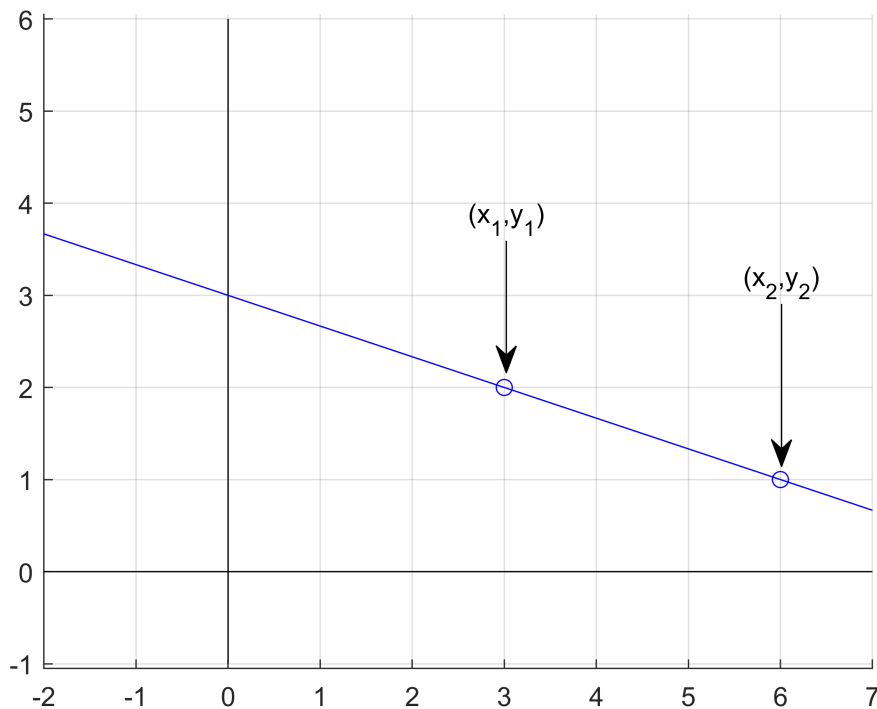
Here is what it looks like in code:

```
% Define the points (x1,y1) and (x2,y2)
x1 = 3;
y1 = 2;
x2 = 6;
y2 = 1;

% Compute the slope
m = (y2-y1)/(x2-x1);

% Compute the y-intercept using m and (x1,y1)
b = y1-m*x1;
```

Here is a plot of the equation of the line we just found:



Determining if a point is on a line

Another part of the problem is determining if a point is on the line.

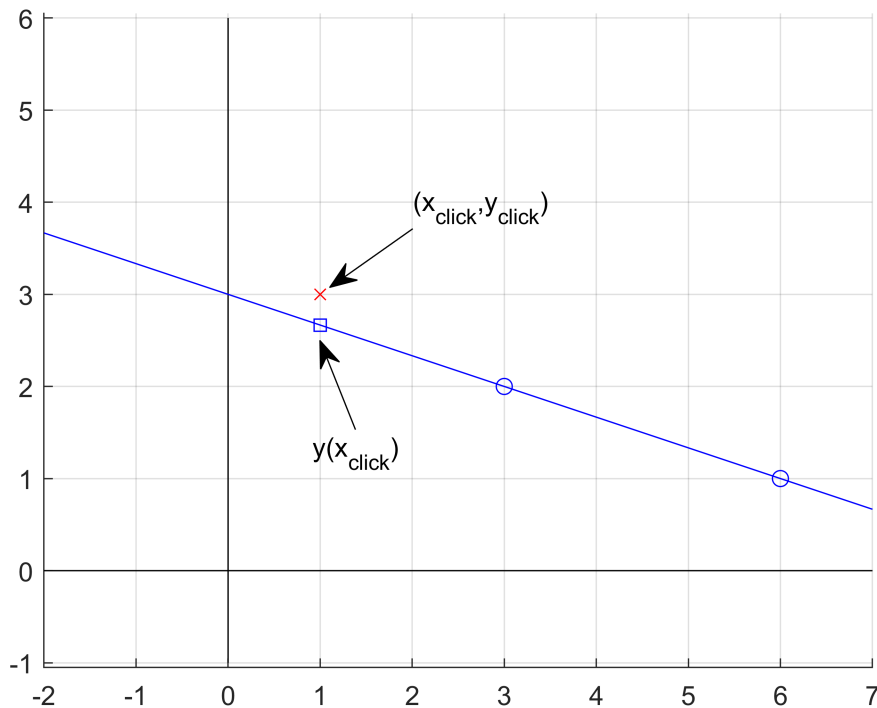
Suppose we want to know if the point (1,3) is on a line.

The best way to do this is to compare the y-value of the point with the y-value that you would get from plugging the x-value into the equation of the line.

Programming Note: Directly comparing an integer (e.g. y-value of the point) with a number that was computed (e.g. the y-value from the line) should not be done.

To get around this, we can round the y-value to say two decimals using MATLAB's round command. For this application, two decimals will not look any different to the user.

The figure below shows that the point (1,3) does not lie on the line we found above.



Here is some code that uses this idea. Assume we already computed the slope, m , and y-intercept, b , from the previous example.

```
% Define the point to test (in your program it will come from ginput)
x_click = 1;
y_click = 3;

% Compute y(x_click) and round it to 2 decimal places
y_at_x_click = round(m*x_click + b, 2);

% Compare y_click to y_at_x_click
if y_at_x_click == y_click
    disp('the clicked point is on the line')
else
```

```
        disp('the clicked point is not on the line')
    end
```

Intersection point of two lines

Let's review how to find the intersection of two lines.

Find the intersection of the following two lines:

$$y = -\frac{1}{3}x + 3 \quad \text{and} \quad y = 2x - 1$$

The idea is simple, set both equations equal to each other (eliminating y) and solve for x .

$$-\frac{1}{3}x + 3 = 2x - 1$$

$$-x + 9 = 6x - 3$$

$$-x - 6x = -3 - 9$$

$$-7x = -12$$

$$x = \frac{12}{7}$$

In your program, you will have to come up with a formula to solve for this x in terms of the slopes and y -intercepts of the two lines.

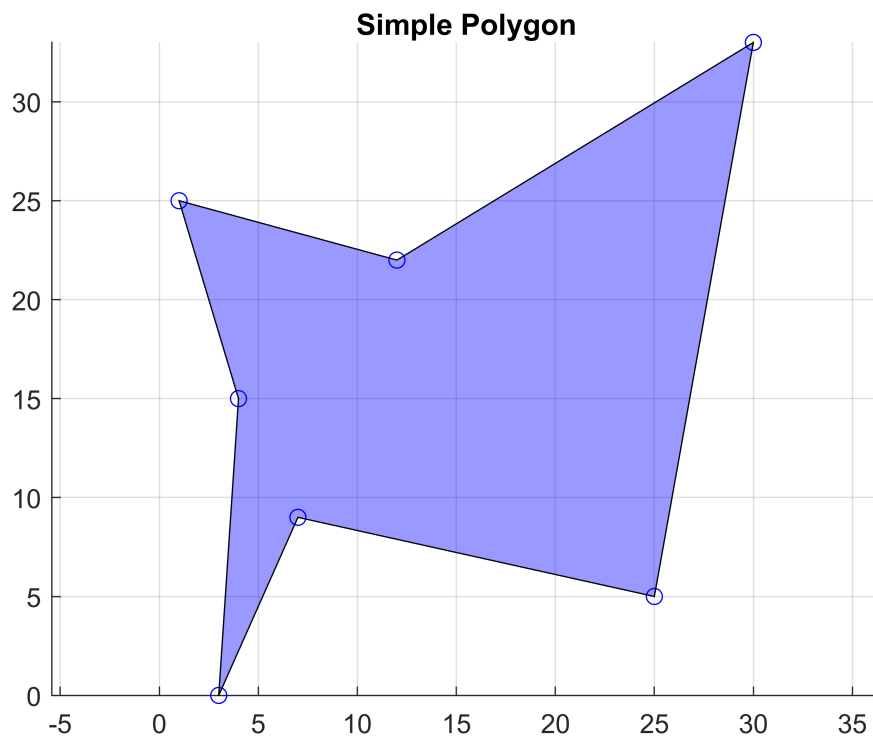
As an exercise, find the formula for the x -value of the intersection point of the following two lines:

$$y = m_1x + b_1 \quad \text{and} \quad y = m_2x + b_2$$

Polygon

A **polygon** is a collection of points (called vertices) that are connected by line segments (called edges).

A **simple polygon** is a polygon that has no intersecting edges. The following polygon is simple:



Here is an example of a polygon that is not simple:

